

0.1 motivation and introduction

0.2 categorics

I want to state and prove the Galois correspondence purely categorically. Throughout fix a ground field k and work in the category of algebraic k -extensions.

Proposition 0.2.1 (A4). For all finite extensions $k \subset K, L$ there exists a diagram:

$$\begin{array}{ccc} & K & \\ \hookrightarrow & & \hookrightarrow \\ k & & \Omega \\ \hookrightarrow & & \hookrightarrow \\ & L & \end{array}$$

Proof. Go by induction on $[K : k]$: the base case has $K = k$ where taking $\Omega = L$ works.

Now suppose $[K : k] > 1$: there will exist some $\alpha \in K \setminus k$, which will have minimal polynomial $f \in k[X]$, let L' be the splitting field of $f \in L[X]$, then

$$\text{Emb}_k(k(\alpha), L') = \{\text{roots of } f \text{ in } L'\} \neq \emptyset$$

and we can form the daigram:

$$\begin{array}{ccc} & K & \\ \hookrightarrow & & \hookrightarrow \\ & k(\alpha) & \\ \hookrightarrow & & \hookrightarrow \\ k & & L' \\ \hookrightarrow & & \hookrightarrow \\ & L & \end{array}$$

where, as $k \neq k(\alpha)$, the tower law says $[K : k(\alpha)] < [K : k]$. Thus, by strong induction, we are done. $\square$

Definition 0.2.2. $k \subset K$ is a **normal extension** if for all $K \subset \Omega$ the right action

$$\text{Emb}_k(K, \Omega) \curvearrowright \text{Aut}_k(K)$$

by **pre-composition** is transitive¹, i.e. every $x : K \hookrightarrow \Omega$ is just the canonical inclusion of $\sigma(K)$ into Ω .

You can never embed a normal extension outside of its image.

Lemma 0.2.3. If $k \subset K$ is normal, and $k \subset F \subset K$ is an intermediate field, then $F \subset K$ is normal.

Proof. Given $x, y \in \text{Emb}_F(K, \Omega) \subseteq \text{Emb}_k(K, \Omega)$ there exists some $\sigma \in \text{Aut}_k(K)$ such that $x = y\sigma$, this σ must be in $\text{Aut}_F(K)$. $\square$

Proposition 0.2.4. Let $k \subset K$ be normal and $k \subset F \subset K \subset \Omega$ be a tower, then the natural restriction:

$$\text{Emb}_k(K, \Omega) \rightarrow \text{Emb}_k(F, \Omega)$$

is surjective, i.e. any embedding $x : F \rightarrow \Omega$ extends to some $\tilde{x} : K \rightarrow \Omega$.

All components of a normal extension present in a bigger field still sit inside a copy of that normal extension.

Proof. For any $x \in \text{Emb}_k(F, \Omega)$, by (A4) I can form the following diagram:

$$\begin{array}{ccccc} & \Omega & \xrightarrow{y} & \Omega' & \\ & \uparrow x & \nwarrow \tilde{x} & \uparrow z & \\ k & \subset & F & \subset & K \end{array}$$

but as $k \subset K$ is normal, I must be able to write $z : K \rightarrow \Omega'$ as $y \circ \sigma$ for some $\sigma \in \text{Aut}_k(K)$. This σ (composed with the inclusion $K \subset \Omega$) is exactly the required $\tilde{x}$. $\square$

¹This action will always be free (no nontrivial element has a fixed point), as all field homomorphisms are injective thus mono. So normal extensions are precisely those extensions $k \subset K$ where $\text{Emb}_k(K, \Omega)$ is an $\text{Aut}_k(K)$ -torsor.

From this, it is immediately obvious that the natural injective composition with $K \subset \Omega$:

$$\text{Emb}_k(F, K) \rightarrow \text{Emb}_k(F, \Omega)$$

is also surjective.

Proposition 0.2.5 (A1). For all extensions $k \subset K, L$ the set $\text{Emb}_k(K, L)$ has cardinality at most $[K : k]$.

Proof. Once again going by induction on $[K : k]$ where the base case is trivial.

Supposing $[K : k] > 1$, pick $\alpha \in K \setminus k$ and consider the restriction map:

$$\rho : \text{Emb}_k(K, L) \rightarrow \text{Emb}_k(k(\alpha), L)$$

Now for any $x \in \text{Emb}_k(k(\alpha), L)$, the pre-image $\rho^{-1}(x) = \text{Emb}_x(K, L)$ has cardinality $\leq [K : k(\alpha)]$, so I can consider the sum:

$$|\text{Emb}_k(K, L)| = \sum_{x \in \text{Emb}_k(k(\alpha), L)} |\rho^{-1}(x)| \leq [k(\alpha) : k][K : k(\alpha)] = [K : k] \quad \square$$

Lemma 0.2.6. For an extension $k \subset K$ and $H \leq \text{Aut}(K/k)$ a finite subgroup, $[K : K^H] \leq |H|$.

Proof. If $G = \{\sigma_1, \dots, \sigma_n\}$, we need to show all sets $\{a_1, \dots, a_{n+1}\} \subset L$ are linearly dependent over K^H . Let $\{a_1, \dots, a_k\}$ consider the vectors:

$$\mathbf{a}_1 = \begin{pmatrix} \sigma_1(a_1) \\ \vdots \\ \sigma_n(a_1) \end{pmatrix}, \dots, \mathbf{a}_{n+1} = \begin{pmatrix} \sigma_1(a_{n+1}) \\ \vdots \\ \sigma_n(a_{n+1}) \end{pmatrix}$$

and wlog let $\{\mathbf{a}_1, \dots, \mathbf{a}_k\}$ be the smallest nontrivial linearly dependent set, so we have $x_i \neq 0$ such that $x_1 \mathbf{a}_1 + \dots + x_k \mathbf{a}_k = 0$, wlog take $x_1 = 1$. Applying some $\sigma \in \text{Aut}(K/k)$ will permute the rows of this linear system, but will always keep x_1 the same, this must always be the same system, otherwise we could subtract and find one of strictly smaller size. Thus all the $x_i \in K^H$ and equation corresponding to $\sigma_i = e$ is as required. $\square$

Corollary 0.2.7 (A5). Under the same assumptions, $[K : K^H] = |H|$, and the obvious inclusion $H \subset \text{Aut}(K/K^H)$ is an isomorphism.

Proof. (A1) already says $|\text{Aut}(K/K^H)| \leq [K : K^H]$, by the previous lemma we know $[K : K^H] \leq |H|$, and by assumption we know $|H| \leq |\text{Aut}(K/K^H)|$. $\square$

Definition 0.2.8. The extension $k \subset K$ is **separable** if for all towers of subfield:

$$k \subset K_1 \subsetneq K_2 \subset K$$

there exists some field Ω with at least two distinct K_1 -embeddings $x, y : K_2 \rightarrow \Omega$.

Lemma 0.2.9. If $k \subset K$ is separable, and $k \subset F \subset K$ is a tower, then $k \subset F$ and $F \subset K$ are separable.

Proof. Any pair of intermediate field of $k \subset F$ or $F \subset K$ are also intermediate field of $k \subset K$. $\square$

Theorem 0.2.10 (Fundamental theorem of Galois theory). If $k \subset K$ is Galois (normal and separable) then there is a bijective, order-reversing map between field and subgroups as follows:

$$H \leq G \mapsto K^H \quad k \subset F \subset K \mapsto \text{Aut}(K/F)$$

Proof. We know the images are fields and subgroups respectively, and the fact that these maps reverse the inclusion ordering is obvious. $H = \text{Aut}(K/K^H)$ is exactly (A5). As all $k \subset F \subset K$ are also Galois, we just have to prove $k' := K^{\text{Gal}(K/k)} = k$ we will go by showing that for all $k \subset \Omega$: $\text{Emb}_k(k', \Omega)$ is just the inclusion $k' \subset K \subset \Omega$, where separability now implies the final result. Consider the tower:

$$k \subset k' \subset K \subset \Omega$$

we know any $\text{Emb}_k(k', \Omega)$ is the restriction of some $\text{Emb}_k(K, \Omega)$ but as $k \subset K$ is normal, this corresponds to composing the inclusion $K \subset \Omega$ with some $\sigma \in \text{Gal}(K/k)$, by construction this is just $k' \subset K \subset \Omega$. $\square$

0.3 translating back down

We want to classify the categorical definitions of normal and separable extensions into field theoretic statements.

Theorem 0.3.1. For a finite extension $k \subset K$ tfae:

1. For all irreducible $f \in k[X]$, either f has no roots in K or f splits completely in K ;²
2. There exists some $f \in k[X]$ such that K is the splitting field of f over k ;
3. $k \subset K$ is normal.

Proof. (1 $\Rightarrow$ 2) If $K = k(a_1, \dots, a_n)$ and f_i is the minimal polynomial of a_i over k respectively, K is clearly the splitting field of $f_1 \cdots f_n$,

(2 $\Rightarrow$ 3) If $k \subset K$ is the splitting field of f with root $a \in K$, then for any $K \subset \Omega$ and $x : K \rightarrow \Omega$, $x(a)$ will always be a root of f in Ω and as K is generated by all the roots of f , $x(K) \subset K \subset \Omega$.

(3 $\Rightarrow$ 1) Now if $k \subset K$ is normal, for any irreducible $f \in k[X]$ let $K \subset \Omega$ be the splitting field of f over K . If f has any root $a \in K$ then we can form the tower $k \subset k(a) \subset K \subset \Omega$ giving the bijection:

$$\{\text{roots of } f \text{ in } \Omega\} = \text{Emb}_k(k(a), \Omega) = \text{Emb}_k(k(a), K) = \{\text{roots of } f \text{ in } K\} \quad \square$$

Lemma 0.3.2. For all $k \subset K$ there exists some $K \subset L$ such that $k \subset L$ is normal.

Proof. Let $K = k(a_1, \dots, a_n)$, with f_i the minimal polynomial of each a_i over k respectively, now let L be the splitting field of $f_1 \cdots f_n$ over K , and thus also over k . $\square$

Definition 0.3.3. For $k \subset K$ define the **separable degree**

$$[K : k]_s := |\text{Emb}_k(K, \Omega)|$$

where $k \subset K \subset \Omega$ is normal extension that exists by the previous lemma.

Lemma 0.3.4. The separable degree $[K : k]_s$ does not depend on the choice of Ω .

Proof. Let Ω_1, Ω_2 be such normal extensions of k , then by (A4) let Ω' be a field containing both of them, we know any embedding $K \rightarrow \Omega'$ must send K to the normal subfield containing it, i.e.

$$\text{Emb}_k(K, \Omega_1) = \text{Emb}_k(K, \Omega) = \text{Emb}_k(K, \Omega_2) \quad \square$$

Separable extensions are now exactly those $k \subset K$ such that for all towers $k \subset K_1 \subset K_2 \subset K$, if $[K_1 : K_2]_s = 1$, then $K_1 = K_2$.

Proposition 0.3.5. For a tower $k \subset K \subset L$ we have

$$[L : k]_s = [L : K]_s [K : k]_s$$

Proof. Take $L \subset \Omega$ such that $k \subset \Omega$ is normal, we know the restriction:

$$\text{Emb}_k(\Omega, \Omega) \rightarrow \text{Emb}_k(L, \Omega) \xrightarrow{\rho} \text{Emb}_k(K, \Omega)$$

is surjective, so we can count

$$[L : k]_s := |\text{Emb}_k(L, \Omega)| = \sum_{x \in \text{Emb}_k(K, \Omega)} |\rho^{-1}(x)| = [K : k]_s [L : K]_s$$

as the previous lemma says the size of $\rho^{-1}(x) = \text{Emb}_K(L, \Omega)$ doesn't depend on the choice $x : K \rightarrow \Omega$. $\square$

Theorem 0.3.6. For a finite extension $k \subset K$ tfae:

- 1.

²This is a standard definition of a normal extension